Counting is matching, construction is multiplication

To count a set X means to find $|X|$. Two master moves solve almost everything: **(A) construct** each object by a sequence of choices (multiply), **(B) match up** X bijectively with a set already counted (inherit its size). Every formula in this chapter is one of these, packaged.

KEY RESULT

Product rule and sum rule (with their exact hypotheses)

Product rule: a task in successive steps, step i offering n_i choices *regardless of earlier choices*, has $n_1 n_2 \cdots n_k$ outcomes. The independence clause is the hypothesis: "5 choices then 4 choices" for choosing 2 people from 5 counts **ordered** pairs (20), because $\{A, B\}$ is built twice.

Sum rule: if X is split into disjoint, exhaustive cases, $|X|$ is the sum of the case counts. Cases must be *every case counted once, every object in some case*.

KEY RESULT

Complement: negatives are rigid, positives are flexible

"At least one" / "no" / "not all" ⇒ count the complement and subtract from the total. Committees of 5 from 12 men and 8 women with at least one woman: $\binom{20}{5} - \binom{12}{5} = 15504 - 792 = 14712$, instead of five direct cases. Strings of length 4 over $\{0,1,2,3,4\}$ containing a 0: $5^4 - 4^4 = 625 - 256 = 369$.

WORKED EXAMPLE

Worked — cases where steps interact

4-digit numbers with distinct digits divisible by 5: split on the units digit (the thousands digit's options depend on it). Units 0: $9 \cdot 8 \cdot 7 = 504$. Units 5: $8 \cdot 8 \cdot 7 = 448$. Total 952. Splitting is the cure when independence fails in exactly one place.

INTUITION

Bijections you already own

Subsets of an n-set ↔ binary strings of length n (in/out per element): 2^n subsets. Increasing functions f with $f(1) < f(2) < f(3)$ into $\{1, \ldots, 5\}$ ↔ 3-subsets: $\binom{5}{3} = 10$. Pairs $A \subseteq B$ of subsets: each element has 3 states (in neither, in B only, in both): 3^n. Monotone/ordered-with-condition almost always means: find the unordered object it encodes.

COMMON TRAP

The three sins of counting (check every answer against them)

1. **Overcounting:** one object built by two choice paths. 2. **Undercounting:** cases not exhaustive. 3. **Broken independence:** multiplying counts that are not fixed per step. Cure for all three: verify a small case by listing. Also ask: objects labelled or unlabelled? order matters or not? is the answer a plausible size (never more than the total pool)?

ASSUMPTIONS & CONVENTIONS

Standing conventions

n, r are non-negative integers with $0 \le r \le n$ unless stated; $\binom{n}{r} = 0$ if $r < 0$ or $r > n$ (a useful convention in sums); $0! = 1$ (exactly one empty arrangement). "Arrangements" are ordered; "selections/committees" are unordered.

Ordered selections

SHORT DERIVATION

nPr from filling labelled roles

Fill r labelled positions from n distinct objects: n options, then $n-1$, ..., then $n-r+1$ (each count fixed regardless of history). Hence

$$P(n,r)=\frac{n!}{(n-r)!}, \qquad P(n,n)=n!$$

Equivalently $P(n,r)$ counts injective functions from an r-set to an n-set; $n!$ counts bijections of an n-set. $0!=1$ is forced: there is exactly one empty arrangement.

KEY RESULT

Identical objects: label, count, divide by the symmetry

Arrangements of a multiset with $n_1,\dots,n_k$ copies of k types ($n=\sum n_i$):

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$

Proof: temporarily label all copies ($n!$ words); each unlabelled word is hit exactly $n_1!\cdots n_k!$ times (permute labels within each type). Division is legal **iff every fibre has the same size** — the exact-symmetry condition; where it fails (necklaces with identical beads), Part II's Burnside lemma is the fix.

Words from COMBINATORICS (C, O, I twice each, 13 letters): $\frac{13!}{2!\,2!\,2!}=778377600$.
MATHEMATICS: $\frac{11!}{2!\,2!\,2!}=4989600$.

WORKED EXAMPLE

Worked — bundles inside identical-letter words

Words from DAAR with the two A's adjacent: bundle AA into one super-object; arrange $\{D, AA, R\}$: $3!=6$ (the identical A's give the bundle one internal order). Words: DAAR, AADR, ADAR, ADRA, ARDA, RADA ✓.

COMMON TRAP

Common traps — permutations

- Dividing by symmetry when the symmetry is not exact (some arrangements fixed by the swap).
- "Choose first, then second" silently counting ordered pairs for an unordered question — off by $r!$.
- Bundles: internal orders multiply only if the bundled objects are distinct; identical objects contribute 1.
- $P(n,r)$ with $r>n$ is 0, not a formula to evaluate.

Around a table

KEY RESULT

Circular permutations: $(n-1)!$

n distinct people around a round table, seatings identified up to rotation: $(n-1)!$. Two proofs: anchor one person as reference and arrange the rest linearly; or divide $n!$ by the n rotations (every orbit has size exactly n for distinct people). If flips are also identified (necklaces of **distinct** beads, $n\geq 3$): $\frac{(n-1)!}{2}$.

Two specified people adjacent (round table): bundle, $2(n-2)!$; not adjacent: $(n-1)!-2(n-2)!=(n-3)(n-2)!$. For 10 people with A, B non-adjacent: $9!-2\cdot 8!=362880-80640=282240$.

The three restriction methods (linear arrangements)

KEY RESULT

Bundle, gap, relative order — pick by the wording

- **"Together/consecutive":** bundle the k objects into one super-object ($k!$ internal orders if distinct), arrange the $n-k+1$ objects.
- **"No two adjacent":** gap method. Arrange the m ordinary objects ($m!$), creating $m+1$ gaps; place the k special objects into **distinct** gaps: $\binom{m+1}{k}$ choices and $k!$ orders: total $m!\binom{m+1}{k}k!$. Six people with A, B, C pairwise non-adjacent: $3! \cdot \binom{4}{3} \cdot 3! = 144$.
- **"Fixed relative order" (A before B before C, not necessarily adjacent):** among all $n!$ orders the k specified objects show each of their $k!$ relative orders equally often: answer $\frac{n!}{k!}$. "A before B" alone: $\frac{n!}{2}$.

WORKED EXAMPLE

Worked — composing methods

Six people P, Q, R, S, T, U in a row; P, Q together and R not at an end. Bundle PQ: $2 \cdot 5! = 240$ arrangements of the 5 objects. Subtract R at an end: 2 ends, then $4!$ arrangements of the rest, bundle order 2: $2 \cdot 4! \cdot 2 = 96$. Answer $240 - 96 = 144$.

Order of operations: bundle first, then apply the positional restriction to the shrunken list.

Alternation: 5 men and 5 women alternating in a row: 2 patterns $\times 5! \cdot 5! = 28800$.

COMMON TRAP

Common traps — restrictions

- Gap method requires the ordinary objects arranged **first**, and needs $k \le m+1$ to be possible at all.
- "Not adjacent" for a specified pair is bundle-and-subtract; "no two of a set adjacent" is the gap method. Swapping them is the classic error.
- Circular "not adjacent" via linear gap method: wrong geometry — use total minus bundled.
- Necklace division by 2 needs distinct beads (or Burnside); identical beads break exactness.

Unordered selections

SHORT DERIVATION

Why dividing by r! is exact here

Every r-subset generates exactly $r!$ orderings, and every ordering comes from one subset: the ordered selections $P(n,r)$ split into equal fibres of size $r!$. So

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}, \qquad \binom{n}{r} = \binom{n}{n-r}$$

The symmetry is complementation: choosing what to keep = choosing what to discard. First question of every problem: do the objects come with positions/labels (permutations) or not (combinations)? Mixed problems do both, in that order.

KEY RESULT

Pascal and hockey-stick (proofs are one split each)

Pascal: $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ — split r-subsets by whether a fixed element is in (then choose $r-1$ more) or out (choose all r from the rest).

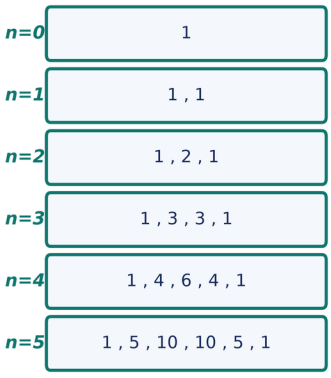
Hockey-stick: $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$ — count $(r+1)$-subsets of $\{1, \ldots, n+1\}$ by their largest element $i+1$: the rest is an r-subset of $\{1, \ldots, i\}$. E.g. $\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \binom{4}{1} = 10 = \binom{5}{2}$ (which is also $1+2+3+4$).

WORKED EXAMPLE

Worked — restricted selections

At least one of 4 special books among 5 chosen from 20: $\binom{20}{5} - \binom{16}{5} = 15504 - 4368 = 11136$ (complement again).

Team of 11 from 7 batsmen, 5 bowlers, 3 all-rounders, at least 3 batsmen and at least 2 bowlers: enumerate by all-rounders $a = 0..3$ and sum products $\binom{7}{b}\binom{5}{w}\binom{3}{a}$ over valid (b, w): $12 + 198 + 660 + 490 = 1360$. The template: one free variable plus bounds makes a finite table — check boundary triples twice, and **do not drop the $\binom{3}{a}$ factor** for the enumerated group (the classic table error: rows for $a = 1, 2$ are each 3 times their unweighted subsums).



Pascal's triangle — row n of (a+b)ⁿ, each entry the sum of the two above

COMMON TRAP

Common traps — combinations

- $\binom{n}{r}$ computed for "arrange the chosen few" problems (or vice versa): decide order first.
- Case splits on "at least" that double-count overlaps — use complement or inclusion–exclusion (sheet 06).
- Boundary triples dropped in group-selection tables (the $(7, 3)$-type corner cases).
- Choosing from groups: multiply the independent group choices, then sum over the valid compositions — not the reverse.

Identical objects into distinct boxes

SHORT DERIVATION

The stars-and-bars bijection

Distributing n **identical** balls into k **distinct** boxes is the same data as a count tuple $(x_1, \ldots, x_k)$, $x_i \geq 0$, $\sum x_i = n$ — and the same as a word of n stars and $k-1$ bars (stars for box 1, bar, stars for box 2, ...). Choosing the bar positions among $n+k-1$ symbols:

$$\binom{n+k-1}{k-1} \quad \text{(empty boxes allowed)}$$

$$\binom{n-1}{k-1} \quad (\text{every box} \geq 1)$$

The positive case: pre-place one ball in each box, distribute the remaining $n-k$ freely. Non-negative solutions of $x+y+z=15$: $\binom{17}{2} = 136$; positive: $\binom{14}{2} = 91$ (difference $45 = 42 + 3$: exactly one zero $3 \cdot 14$, exactly two zeros 3 ✓).

KEY RESULT

The same formula in four costumes

- **Multisets:** choosing 12 fruits from 5 varieties (counts only): $\binom{12+5-1}{4} = \binom{16}{4} = 1820$.
- **Compositions:** ordered sums of n into k positive parts: $\binom{n-1}{k-1}$ (separators in the $n-1$ gaps between unit blocks); any number of parts: 2^{n-1}. Ordered sums of 10 into parts ≥ 2: shift $b_i = a_i - 1$, sum over the number of parts $k = 1..5$: $1 + 7 + 15 + 10 + 1 = 34$ (the $k = 1$ term is the single-part sum $[10]$ itself — easy to drop).
- **Non-negative solutions with lower bounds:** shift each variable down to 0 first.
- **Upper bounds:** complement/inclusion–exclusion: $x_1 + x_2 + x_3 + x_4 = 20$, $x_1 \leq 5$: total $\binom{23}{3} = 1771$ minus ($x_1 \geq 6$, shift) $\binom{17}{3} = 680$ gives 1091.

COMMON TRAP

Common traps — stars and bars

- Identical vs distinct balls: distinct balls into boxes is k^n (functions), a completely different count. Read which objects are identical.
- Applying $\binom{n+k-1}{k-1}$ to ordered compositions with positive parts (that is $\binom{n-1}{k-1}$).
- Upper-bound problems with two violated bounds need genuine inclusion–exclusion, not one subtraction.
- Balls identical + boxes identical (partitions) has **no closed form** — Part II; do not invent one.

The principle

SHORT DERIVATION

Inclusion–exclusion, proved by tracking one element

For bad events $A_1, \ldots, A_n$ in a universe X:

$$\left|X \setminus \bigcup A_i\right| = \sum_{S \subseteq \{1..n\}} (-1)^{|S|} \left|\bigcap_{i \in S} A_i\right|$$

Proof: an element lying in exactly $t \geq 1$ of the sets is counted $\sum_{j=0}^{t} (-1)^j \binom{t}{j} = (1-1)^t = 0$ times; an element in none is counted once. **Universal recipe:** define bad events; compute the intersection size for a generic set of k specified events (usually $\binom{n}{k} \times$ something free); write the alternating sum.

Integers 1–1000 divisible by 2, 3 or 5: $500 + 333 + 200 - 166 - 100 - 66 + 33 =$ (so divisible by none: 266).

KEY RESULT

Derangements: the flagship application

D_n = permutations of $\{1..n\}$ with **no** fixed point. With A_i = "i is fixed", a specified set of k fixed points leaves $(n-k)!$ free permutations:

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

$D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 =$. Also $D_n = (n-1)(D_{n-1} + D_{n-2})$ (where does element 1 go? then whether its partner swaps back), and D_n is the **nearest integer to** $n!/e$ since the tail beyond n is $< \frac{1}{n+1} < \frac{1}{2}$.

Exactly k fixed points: $\binom{n}{k} D_{n-k}$ (for $n = 5$, exactly two correct letters: $\binom{5}{2} \cdot 2 = 20$; check: $44 + 20 + 45 + 10 + 0 + 1 = 120 = 5!$ ✓).

WORKED EXAMPLE

Worked — onto functions (IE in disguise)

Onto functions from a 5-set to a 3-set: total 3^5 minus those missing a value: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$.

Hands of 6 cards containing all four suits: $\binom{52}{6} - 4\binom{39}{6} + 6\binom{26}{6} - 4\binom{13}{6} = 8682544$.

COMMON TRAP

Common traps — inclusion–exclusion

- Sign alternation starting wrong: the empty intersection (whole universe) carries +.
- Intersection sizes computed for *exactly* k events instead of *at least these* k (IE wants the latter).
- Bounded stars-and-bars terms with negative top: $\binom{m}{k-1}$ for a negative-sum equation is **0** — terminate the sum, never evaluate.
- Derangement problems with only *some* positions forbidden: restrict the event set (then it is partial derangement / rook theory, Part II flavour).

Existence by averaging

KEY RESULT

Pigeonhole principle (the art is the boxes)

$n+1$ objects in n boxes $\Rightarrow$ some box has ≥ 2. The theorem is trivial; every application is the choice of the box map so that "same box" gives the conclusion.

Odd-part box: from any $n+1$ integers in $\{1, \ldots, 2n\}$, two share the same odd part in $x = 2^a \cdot m$ (m odd) — only n odd parts exist — so one divides the other. Sharp: $\{n+1, \ldots, 2n\}$ has no such pair.

Geometry box: 5 points in a unit square: quarter it into 4 subsquares; two points share one, at distance $\leq \frac{\sqrt{2}}{2}$ (its diagonal).

Counting by last-step decomposition

KEY RESULT

Recurrences: split by the last block

If objects of size n end with one of a few block types, leaving a smaller instance of the same problem, the counts satisfy a recurrence. $2 \times n$ domino tilings: ends vertical (a_{n-1}) or two horizontal (a_{n-2}): $a_n = a_{n-1} + a_{n-2}$, $a_0 = a_1 = 1$, so $a_n = F_{n+1}$ (tilings of 2×10 : 89). Binary strings with no consecutive 1s and subsets with no two consecutive give the same recurrence (F_{n+2}). Staircase with steps 1–2: $a_{10} = 89$; steps 1–3: tribonacci. Guard the base cases — a correct recurrence with wrong seeds is a wrong answer.

KEY RESULT

Recap — the chapter on one card

Product/sum/complement; bijection = same count. $P(n,r) = \frac{n!}{(n-r)!}$; identical objects $\frac{n!}{\prod n_i!}$; circular $(n-1)!$ (necklace $\frac{(n-1)!}{2}$ distinct beads). Bundle / gap $m!\binom{m+1}{k}k!$ / relative order $\frac{n!}{k!}$. $\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$; Pascal; hockey-stick. Stars and bars $\binom{n+k-1}{k-1}$ / positive $\binom{n-1}{k-1}$; compositions 2^{n-1}. IE alternating sum; $D_n = n!\sum\frac{(-1)^k}{k!} \approx n!/e$; onto $= \sum(-1)^j\binom{k}{j}(k-j)^n$. Pigeonhole: design the boxes. Recurrence: split by last block, mind the seeds.

COMMON TRAP

Pre-submission checklist

Order matters? Labels? Identical objects — is my division by symmetry exact (equal fibres)? Cases disjoint and exhaustive? Did "at least" get complemented? Bounds handled by shift/IE, with negative-top terms set to 0? Small case listed by hand? Answer size sane against the total pool?

INTUITION

Forward map

Binomial identities are this chapter counted twice (chapter 9); lattice paths and Catalan, Burnside–Pólya, partitions and Stirling numbers, intersecting families: Part II here. The through-line never changes: construct, split, or match up.